2022 IEEE 2nd International Conference on Information Communication and Software Engineering (ICICSE) | 978-1-6654-8220-2/22/$31.00 ©2022 IEEE | DOI: 10.1109/ICICSE55337.2022.9828915

# Image Matching Based on Fast PCA-SIFT Descriptors with Automatic Determination of Dimensionality for PCA

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***Abstract*—Image matching is a key technology in the field of image processing. An effective image matching method based on fast PCA-SIFT descriptors is proposed and studied deeply. Firstly, the matrix left multiplication method is used to reduce the operation load of PCA and improve its operation speed. Secondly, we utilize the total interpretation proportion of the data variance of the top several principal components to determine the optimal dimension of the descriptor for PCA, thus the number of retained principal components can be determined automatically. Some intuitive and persuasive simulation experiments are carried out by using the proposed method. Experimental results demonstrate that the proposed method can automatically determine the number of retained principal components, and can reduce the operation load of PCA. The proposed image matching method can be used in the fields of three-dimensional reconstruction, cooperative augmented reality and teleoperation robots.**

***Keywords—image processing, image matching, SIFT, PCA, dimensionality reduction, automatic determination of dimensionality***

## I. Introduction

Image matching based on feature point correspondence is the core technology of multiple view scene reconstruction, and it is still a research focus at present [1-3]. It usually includes the steps of feature point detection, feature description and feature matching. It needs to detect the stable feature points and determine the affine transformation parameters such as their position, direction and scale of these feature points in the image. The scale-invariant feature transform (abbreviated as SIFT) based on difference-of-Gaussian (abbreviated as DoG) images is one of the current feature point detection algorithms with good performance [4-5]. However, SIFT descriptors result in feature vectors with 128 dimensions. Under the existing computer performance conditions, SIFT algorithm is difficult to have real-time performance without the help of hardware acceleration. For the teleoperation robots based on augmented reality, it will produce time delay, reduce the sense of telepresence, and even lead to misoperation.

Dimensionality reduction of descriptors is a method worthy of trial and in-depth research. Speeded up robust features (abbreviated as SURF) descriptors based on integral image only generate 64 dimensional feature vectors [6-7]. The SURF algorithm is fast and has good performance as the same as the SIFT algorithm, but it is not stable to rotation and illumination changes [8]. In 2004, Ke and Sukthankar proposed a novel local feature descriptor, namely a PCA-SIFT descriptor [9]. It uses principal component analysis (abbreviated as PCA) instead of direction histograms to construct descriptors. Through many experiments, it is known that the PCA-SIFT descriptor achieves the optimal matching performance when the dimensionality of descriptor is 36. However, the optimal dimension for PCA cannot be determined automatically in the method. This is one of its disadvantages. Most of the operation load of PCA is to solve the eigenvalues and eigenvectors of covariance matrix of the sample data. When the dimensionality of covariance matrix is high, the operation will be very time-consuming, and even the memory will be exhausted. This is another disadvantage of it. To solve the above problems, this paper proposes an improved PCA-SIFT descriptor for image matching. This improved algorithm can not only increase the operation speed of PCA, but also automatically determine the optimal dimensionality of descriptor.

The remainder of the paper is organized as follows. First, the SIFT algorithm and the PCA-SIFT algorithm are reviewed briefly in Section II. In Section III, a fast PCA-SIFT descriptor is introduced, and its acceleration principle is presented in detail. A method that can automatically determine the optimal dimension of the descriptor for PCA is shown in detail in Section IV. Some intuitive and persuasive simulation experiments are presented in Section V. Section VI provides some conclusions and future research directions.

## II. Review of SIFT and PCA-SIFT

In this section, the feature point detection and feature point descriptor based on SIFT, and the PCA-SIFT descriptor will be briefly reviewed.

### *A. Feature Point Detection Based on SIFT*

Feature point detection based on SIFT is to find extreme points in scale space, and then detect stable feature points. It can deal with the changes of illumination, scale and rotation, and has certain affine invariance.

Firstly, the positions and scales of feature points need to be preliminarily determined. In scale space, some useful features at different resolutions can be extracted. Assuming that there is a two-dimensional image $I(x,y)$, and the scale space of $I(x,y)$ is defined as

$$L(x,y,\sigma)=G(x,y,\sigma)*I(x,y) \tag{1}$$

where $G(x,y,\sigma)$ is a Gaussian function with variable scale $\sigma$, as shown below

$$G(x,y,\sigma)=\frac{1}{2\pi\sigma^2 e^{-\left(x^2+y^2\right)/2\sigma^2}} \tag{2}$$

Large scale features correspond to the general features of the image, and small scale features correspond to the detailed features of the image.

The original image is continuously sampled in reduced order to obtain a series of images of different sizes. These images build the pyramid model of the image from large to small and from bottom to top. The number of layers of the pyramid is jointly determined by the size of the original image and the size of the image on the top of the pyramid. Then, the images of each layer of the pyramid are smoothed using Gaussian functions with different parameters.

The next feature point detection is carried out in scale space of DoG, which is defined as follows

$$D(x,y,\sigma)=L(x,y,k\sigma)-L(x,y,\sigma) \tag{3}$$

where $k$ is a scaling factor.

The preliminary detection of feature points is completed by comparing the images of adjacent two layers in the same octave of DoG pyramid. Next, the precise positions and scales of feature points need to be determined.

The scale space is regarded as a surface. According to the gradient of the gray value of the pixels in the neighborhood of the feature point, a three-dimensional quadratic function is fitted to determine the sub-pixel position and scale of the feature point precisely.

### B. SIFT Descriptors

After obtaining the position of feature points, it is necessary to describe the feature points uniformly, so that the descriptors have rotation invariance. Then, through the matching of descriptors, the corresponding relationship of feature points is established. SIFT descriptors have good properties such as illumination invariance, rotation invariance and scale invariance.

Firstly, the gradient of pixels in the neighborhood of feature points is calculated, including the moduli and orientations of gradient. Then, a histogram is used to count the moduli and orientations of gradient of pixels in the neighborhood of feature points. The peak of the histogram corresponds to the gradient orientation of the feature point. And the orientation is used as the dominant orientation of the feature point. In this way, the descriptors have rotation invariance.

In the above solution, the size of the neighborhood used is $16\times16$ pixels. Therefore, SIFT descriptors are feature vectors with 128 dimensions.

### C. PCA-SIFT Descriptors

In order to reduce the dimensionality of SIFT descriptor, Ke and Sukthankar proposed PCA-SIFT descriptors in 2004 [9]. PCA-SIFT descriptors have better illumination invariance than those of SIFT descriptors [8]. In [9], the PCA-SIFT descriptor only uses the classical PCA method instead of orientation histograms to generate low-dimensional feature descriptors. Other operations, such as feature point detection and determination of the dominant orientation of feature points, are the same as those of SIFT descriptors.

First, a patch with a size of $41 \times 41$ pixels around the feature point is selected, and is rotated to the dominant orientation of the feature point. Then, gradients in vertical and horizontal directions of the patch are calculated to form a vector with $39 \times 39 \times 2 = 3042$ dimensions. Let $n$ be the number of feature points. Thus, an $n \times 3042$ matrix $A$ is constructed. Next, the matrix $A$ is zero-centered to obtain a zero-centralization matrix $B$. Finally, the covariance matrix of the zero-centralization matrix is obtained, and the eigenvalues and eigenvectors of the covariance matrix are further solved.

Suppose that the dimensionality of the eigenvectors needs to be reduced to $k$ dimensions. The top $k$ eigenvectors are selected to form a $k \times 3042$ projection matrix $C$. Subsequently, the dimensionality of the eigenvector can be reduced from 3042 to $k$ by multiplying the zero-centralization matrix $B$ and the projection matrix $C$.

## III. Fast PCA-SIFT Descriptors

PCA is a multivariate analysis method, which is often used to reduce the dimensionality of feature vectors and handle the curse of dimensionality. In essence, it projects the sample data in high-dimensional space into low-dimensional space through linear transformation on the premise of representing the original data as well as possible.

In the PCA algorithm, it is very time-consuming to solve those eigenvalues and eigenvectors of the covariance matrix of samples. For instance, for the projection matrix $C$ mentioned earlier, its covariance matrix will be a $3042 \times 3042$ square matrix. Therefore, when the dimensionality of feature vector is high, the operation load will be very heavy, and even the computer memory will be exhausted. A fast PCA method can be utilized to solve the eigenvectors corresponding to the non-zero eigenvalues of the covariance matrix with less operation load [10].

Suppose there is an $n \times d$ sample matrix $X$, $n$ is the number of samples, and $d$ is the dimensionality of feature vector of each sample. In general, $n$ is far less than $d$. The sample matrix $X$ is zero-centered, and a zero-centralization matrix $Z$ can be obtained. Then its covariance matrix $S$ is a $d \times d$ matrix, which is equal to the matrix $Z^{\mathrm{T}} Z$ .

Now, suppose $R$ is a matrix of size $n \times n$ , and make it equal to the matrix $ZZ^{\mathrm{T}}$ . Comparing the matrix $R$ and the matrix $S$ , it can be seen that the size of $R$ is much smaller than that of $S$ . In this way, it will be more convenient to solve those eigenvalues of $R$ . Moreover, $S$ and $R$ have the same non-zero eigenvalues. Therefore, the eigenvalues of $S$ can be easily obtained by solving the eigenvalues of $R$ . A concise derivation is shown below.

Assuming that an $n$ -dimensional column vector $v$ is the eigenvector of the matrix $R$ , there is

$$\left(ZZ^{\mathrm{T}}\right)v = \lambda v \tag{4}$$

Both sides of the above equation are left multiplied by the matrix $Z^{\mathrm{T}}$ at the same time, as follows

$$\left(Z^{\mathrm{T}}Z\right)\left(Z^{\mathrm{T}}v\right) = \lambda\left(Z^{\mathrm{T}}v\right) \tag{5}$$

Since the matrix $S = Z^{\mathrm{T}}Z$ , the eigenvalues of $S$ are $Z^{\mathrm{T}}v$ .

Therefore, the eigenvalues and eigenvectors of large matrix $S$ can be obtained indirectly by solving the eigenvalues and eigenvectors of small matrix $R$ . Because the size of $R$ is much smaller than that of $S$ , this method can reduce the operation load of classical PCA algorithm and improve the operation speed.

## IV. Automatic Determination of Dimensionality for PCA

The available information used to distinguish feature points is restricted by the maximum dimension of the feature vectors of feature points extracted by SIFT descriptors. Since the principal components come from the multidimensional feature vectors of feature points, the maximum number of principal components is equal to the maximum dimension of the eigenvectors.

In feature dimensionality reduction, only some principal components are retained. Generally, the first principal component contains the most data variance, followed by the second principal component, and the variance degree of the subsequent principal components decreases in turn.

At present, there are three common methods to determine how many principal components are retained.

The first method is based on those eigenvalues of principal components. If the eigenvalue is less than a certain threshold, it is considered that the principal component has little explanation for the data variance and it should be eliminated. The defect of this method is that if those eigenvalues of some principal components are very close to the threshold, the hint of this method on the number of retained principal components becomes less obvious. At this time, other methods need to be used to assist in the determination.

The second method is based on the proportion of interpretation of data variance. This method has two optional parameters. One is the interpretation proportion of the data variance of a single principal component, the other is the total interpretation proportion of the data variance of the top several principal components. Here, we use the second parameter.

Let $E_k$ be the total interpretation proportion of the data variance of the top $k$ principal components, and it is defined as follows

$$E_k = \frac{\sum_{i=1}^{k} e_i}{\sum_{i=1}^{N} e_i} \tag{6}$$

where $e_i$ is the eigenvalue of each principal component, $k$ is the number of retained principal components, and $N$ is the total number of principal components.

Previous studies have suggested that, the retained single principal component should be able to explain at least 5-10% of the data variance, and the top several retained principal components should be able to explain 60-70% of the data variance. It is obvious that the method is subjective. If the percentage values used are different, the number of retained principal components will often be different.

The third method is to utilize a scree plot to determine how many principal components should be retained. The scree plot can reflect the degree of interpretation of data variance by each principal component. In the scree plot, the abscissa corresponds to the index numbers of the principal components, and the ordinate can correspond to eigenvalues or percentages of data divergence. The number of retained principal components can be determined by the position of steep slope tends to be gentle.

It can be seen from the above that the number of principal components retained by different methods is not exactly the same. This needs to make a choice according to the research experience and purposes. In short, determining the number of retained principal components is a subjective decision, and there is no optimal method. The advantages and disadvantages of these three methods are relative.

## V. Experiments and Results

After a detailed introduction of the proposed method, experimental results will be presented in the section. In order to test the effectiveness of the proposed method, we have carried out some intuitive and persuasive simulation experiments.

We intercept several small-size images with overlapping areas from a large-size mountain image [11] to build an image set [12]. From this image set, two images of mountains are selected for image matching experiment. Fig. 1 (a) is a reference image, and Fig. 1 (b) is an image to be matched.

(a) (b)

Fig. 1. Two images of mountains with overlapping areas to be matched.

Next, the SIFT method and the proposed method are used for image matching respectively. The feature point matching results of the two methods are shown in Fig. 2. Fig. 2 (a) and Fig. 2 (b) are the results of coarse matching and fine matching using the SIFT method respectively. Fig. 2 (c) and Fig. 2 (d) are the results of coarse matching and fine matching using the proposed method respectively. In the fine matching, a random sample consensus (abbreviated as RANSAC) method [13] is used, which is a classical method for removing outliers. In the RANSAC method, firstly, four pairs of matching points that are not on the same line are randomly selected, and then a projective transformation matrix is obtained. For each matching point transformed by the projective transformation matrix, the distance from it to the corresponding matching point is calculated. If the distance is less than a user-defined threshold, the point is considered an interior point and retained. Otherwise, it will be treated as an outlier and rejected.

Both methods have four steps, namely, feature point detection, determination of the dominant orientation of feature points, descriptor generation and feature point matching. These two methods are only different in the step of descriptor generation.

In feature point detection, the number of feature points detected in the reference image is 52, and the number of feature points detected in the image to be matched is 108. The size of feature matrices corresponding to the two images, as well as the number of point pairs of coarse matching and fine matching by using the two methods respectively, are shown in Table I. In the proposed method, the number of retained dimensions is set to 30. Although the number of point pairs of the proposed method in the fine matching is 6, which is less than that of the SIFT method in the fine matching, the stitched images of these two methods are almost the same. For the sake of brevity, these two stitched images are not shown.

(a) Coarse matching of image pairs using SIFT

(b) Fine matching of image pairs using SIFT

(c) Coarse matching of image pairs using the proposed method

(d) Fine matching of image pairs using the proposed method

Fig. 2. Results of image matching using the SIFT method and the proposed method.

TABLE I. QUANTITATIVE EVALUATION OF IMAGE MACHING METHODS

| Image matching methods | Size of feature matrix of reference image | Size of feature matrix of image to be matched | Number of coarse matching point pairs | Number of fine matching point pairs |
|---|---|---|---|---|
| SIFT | 52×128 | 108×128 | 29 | 12 |
| The proposed method | 52×30 | 108×30 | 21 | 6 |

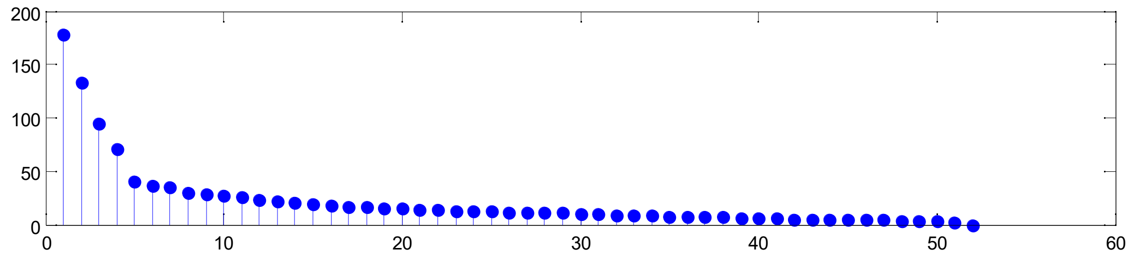


Fig. 3. Eigenvalues obtained from the feature points in the reference image.

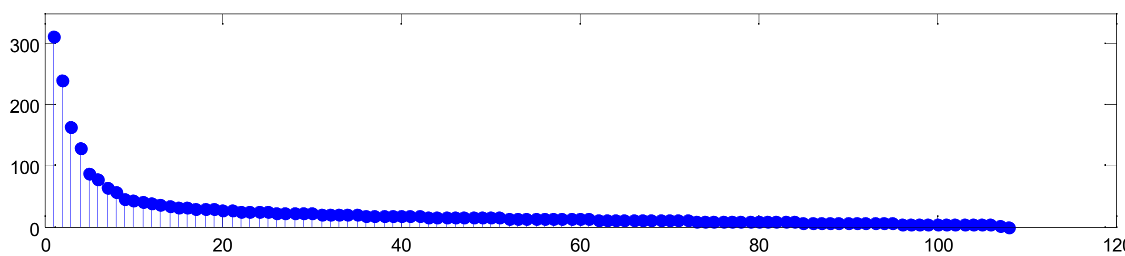


Fig. 4. Eigenvalues obtained from the feature points in the image to be matched.

TABLE II. DETAILED INTERPRETATION PROPORTIONS OF THE DATA VARIANCE OF THE TOP K PRINCIPAL COMPONENTS

| | $k = 27$ | $k = 28$ | $k = 29$ | $k = 30$ | $k = 31$ | $k = 32$ |
|---|---|---|---|---|---|---|
| Reference image | 86.01% | 87.00% | 87.96% | 88.86% | 89.74% | 90.55% |
| Image to be matched | 66.37% | 67.22% | 68.05% | 68.87% | 69.65% | 70.43% |

It can be seen from Table I that the sizes of the feature matrices of the proposed method is about one quarter of those of the SIFT method. In this way, the dimension number of principal components can be reduced.

In the PCA-SIFT method, the feature vectors of the reference image and the image to be matched are both in 3042 dimensions. This dimensionality is a fixed positive integer.

However, in the proposed method, the retained dimensions are no longer fixed positive integers, and they will change with the scenes. To be exact, they are determined by the number of feature points extracted in the scene images. In this experiment, the maximum dimension of the eigenvector in the reference image is equal to the number of extracted feature points, namely, 52 dimensions. Similarly, the maximum dimension of the feature vector in the image to be matched is equal to 108. This is because the principal components come from the multidimensional feature vectors of feature points, and the maximum number of principal components is equal to the maximum dimension of the feature vectors. When performing the principal component analysis on the feature points in the reference image, 52 eigenvalues can be obtained, as shown in Fig. 3. Similarly, when performing the principal component analysis on the feature points in the image to be matched, 108 eigenvalues can be obtained, as shown in Fig. 4. These eigenvalues are all sorted in descending order, and they are clearly shown in Fig. 3 and Fig. 4.

For image matching, the maximum number of retained principal components is required not to be greater than the smaller number of feature points detected in the reference image and the image to be matched. In other words, the maximum number is required to be no more than 52.

Next, we use the second method described in section IV to determine the minimum number of retained principal components. We can utilize the total interpretation proportion $E_k$ of the data variance of the top $k$ principal components to determine the optimal dimension of the descriptor for PCA, as shown in (6). The total interpretation proportion of the reference image is 1111.2, and that of the image to be matched is 2572.1. The detailed interpretation proportions of the data variance of the top $k$ principal components are shown in Table II.

The experimental results show that for this kind of mountain images, when $k$ is less than 30, the image matching fails. Thus, the minimum number of retained principal components is 30. According to Table II, the threshold of interpretation proportion can be set to 68.87%. Therefore, it can be concluded that for such mountain scenes, setting the threshold of interpretation proportion to 70% can automatically determine the number of retained principal components.

## VI. Conclusions and Future Research Directions

In this paper, an effective image matching method based on fast PCA-SIFT descriptors is proposed and studied in depth. And the proposed method can automatically determine the dimensionality for PCA. Simulation experiments have been carried out by using the proposed image matching method. Those experimental results show that the proposed image matching method can match feature point pairs accurately and effectively, and save more memory.

However, the PCA method has several limitations. For example, each principal component is required to be orthogonal, and the divergence of retained features is expected to be the maximum. For these limitations, an alternative method is to utilize an independent component analysis (abbreviated as ICA) method [14-15]. It does not require the orthogonal relationship among its components, and it does not need to consider the divergence of features when determining the retained components. A compressed sensing method can also be used for feature dimensionality reduction [16-17], and thus it is also worthy of in-depth research.

## Acknowledgment

This work was supported in part by the National Natural Science Foundation of China under Grant No. 61472227, No. 61806006 and No. 61801269, in part by the Project of Shandong Province Higher Educational Science and Technology Program under Grant No. J14LN02 and No. J18KA365, in part by the Key Research and Development Project for Integration of Universities, Institutes and Urban Industries in Yantai under Grant No. 2018XSCC035, in part by the Project of Shandong Provincial Key Research and Development Program (Science and Technology Key Project of Public Welfare) under Grant No. 2019GGX101071, and in part by the Doctoral Foundation of Anhui University of Science and Technology under Grant No. ZY543.

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